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Mohali, Punjab
India

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Team Kalki: Preliminary Design Review

1. Leadership and Team Structure

Leadership: Plaksha University's Team Kalki is composed of six subsystems, each having a designated lead who acts as the link between their team and other subsystems. Dr. Deepan Muthirayan is our team's faculty mentor and Manan Gadesha leads our team as the Chief Executive Officer. Our team comprises 50 student members, who belong to Robotics, Bioengineering, CS-AI and DSE majors. The member selection criteria involved 200+ interviews, a month-long training period, and a final assessment, with member experience ranging from first to third year students. All subsystem heads have extensive previous experience in similar projects. Our outreach includes conducting demonstrations during events, and our team is currently working to secure academic credits for the project to encourage broader student participation and involvement.

Communications: Each subsystem includes at least two members that work across multiple subsystems. Team leads meet 3-4 times a week, with complete subsystem meetings held bi-weekly and full team meetings once a month. Apart from physical meetings, the team uses a WhatsApp community with separate groups for each subsystem along with a unified URC group for overall communication. Our team uses Google Drive as the primary file sharing platform for documentation, GitHub for code sharing, and a shared Fusion 360 hub for CAD collaboration.

2. Team Resources

Finances: The team received funds from Plaksha Robotics Club, private donations and industry sponsorships from IndiaMART and Rototon Polypack. We are actively pursuing sponsorships from various corporate companies and research partners. The team has been allotted a budget code (official bank account) by the university to keep all official deposits consolidated. Each week the subsystem heads provide the CFO with a BOM to procure parts essential for developing the rover according to priority.

Facilities: All the members of Team Kalki have 24/7 access to Plaksha University's robotics lab with dedicated workstations in the lab for all subsystem heads and plenty of space for the project, ensuring optimal design and inventory management. The Makerspace helps us in manufacturing mechanical parts of the rover through CNC, laser cutting, etc. We have access to multiple high-tech 3D printers, woodworking tools, etc. The team has been allocated land to build a testing site.

3. Project Management Plan

Development Lifecycle Approach: The team uses an iterative development approach. Two rovers, V1 and V2, are planned to be created. The V1 rover serves as a software testbed and a platform for electrical and mechanical learning and development. The project management team handles deadlines and planning. Subsystems work collaboratively in parallel, integrating design ideologies and gradually increasing the rover's complexity. The management team supports and monitors progress, and subsystem heads meet 4 times a week.

Systems Integration: Individual components are designed and fabricated, and through cross-integration incremental steps are taken so that the parallel work can be collaborative. Simulations and simple test cases are designed to test each new component, and major checkpoints are tested on the V1 rover, including the

autonomous navigation algorithms. Once tested, they are integrated into the final system and the development cycle progresses. Once the V2 rover is finalized, continuous testing and training sessions will be conducted, using on-campus sites to create mock landers and testbeds. These trials will aid in the refinement process, and sequentially as more and more complex tasks are shown, the entire mission will gradually be integrated and trained.

4. Preliminary Technical Design

Our V1 rover prototype features a 600 mm wheelbase built around a main body, rocker-bogie suspension, and an under-mounted differential pivot. The system supports 6-wheel Ackerman and differential steering for point turns, while the pivot enables dynamic weight transfer for improved stability and traction on rough terrain. The rocker-bogie mechanism maintains continuous wheel-ground contact for reliable odometry, and custom TPU honeycomb tires provide optimal compression and grip in complex environments.

Extensive testing of V1 has validated our mobility and drivetrain approach for V2, which integrates a robotic arm for the Delivery and Servicing mission. The V2 rover will employ a 6-DoF manipulator, with each joint incorporating an encoded stepper-driven cycloidal gearbox for high torque density and low backlash. Direct joint actuation extends the reach envelope, while joint encoders enable closed-loop, repeatable precision for high-accuracy tasks.

Power is supplied by two custom 36 V (10S) Li-ion batteries in parallel with a built-in BMS. A custom Power Distribution Board with DC-DC converters, sensors, and an emergency stop delivers regulated power to all subsystems. Both rocker bogies include their own PCBs and PDBs to reduce wiring between the bogies and the main body.

The electronics are organized into three nodes: actuator, sensor, and communication, linked redundantly through a master Raspberry Pi 5 and an NVIDIA Jetson. The Jetson bridges high- and low-level software, while the Pi 5 manages communication between the rover, arm, and biosciences module. Nodes communicate via CAN-bus and USB/UART. GPS and 3D LiDAR support localization and path planning, and communication uses a 900 MHz link for manual override, 2.4 GHz MAVLink telemetry, and 5.8 GHz video transmission with Yagi, cloverleaf, and dipole antennas.

The software architecture uses a "Heterogeneous Cluster" design. An NVIDIA Jetson T5000 handles 3D LiDAR processing and global scan matching, coordinated by the Pi 5, which manages dedicated nodes for sensor fusion, actuation kinematics, and communication. Navigation replaces full SLAM with a "Truth Verification" system: a custom D* algorithm plans routes on pre-loaded USGS terrain maps, and periodic terrain scans reset global position to eliminate drift. A meta-monitor called the "Idea of Truth" continuously validates data integrity through IMUs and encoders.

For equipment servicing, a YOLO v11 pipeline with RGB-D correlation provides precise 3D object localization, aided by an arm-mounted camera. We have already implemented autonomous keyboard typing and semi-autonomous USB drive insertion using an Orangewood arm for testing, to be replaced with the mission-ready manipulator. Low-level control relies on distributed Raspberry Pi Pico nodes on a hybrid communication bus, enabling synchronized round-robin scheduling and adaptive feedback loops where real-time RPM checks tune PID parameters. The current V1 implementation uses a simpler I^2C-based architecture with point-turn and differential-drive control. V2 will incorporate custom overrides, structured data-flow design, and complete closed-loop feedback, observable in real-time.

V1's Soil Acquisition System uses a high-torque Archimedes' screw in a stationary shroud to extract subsurface regolith while preventing fallback. Samples are partitioned into one cached aliquot and three aliquots routed for in-situ testing. Environmental sensors measure temperature, humidity, pressure, and soil pH. Bio-detection mixes regolith with sterile water, filters it through a 0.45 µm membrane, and analyzes the supernatant with a MEMS micro-spectrometer for Ninhydrin (amino acids) and Bradford (proteins) as-

says. V2 adds a heated Benedict's test for reducing sugars, with the spectrometer verifying optical-density shifts across all three assays for robust biochemical redundancy.

Graphics and Tables

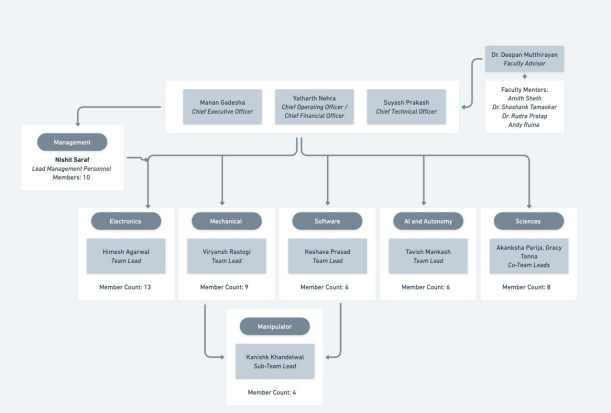

Figure 1: Organizational Chart

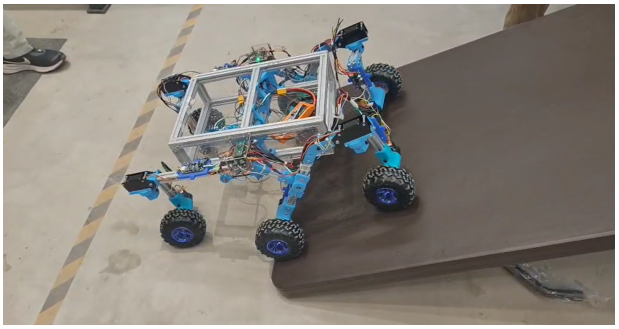

Figure 2: V1 Rover Climbing a Table

INCOME		
Actual Income to Date		
Plaksha Robotics Club	6400	
Plaksha University	15350	
Total Income Received to Date	**21750**	
Anticipated Income		
IndiaMART		12000
Private Funding		5000
Rototon Polypack Pvt. Ltd.		1300
Total Income Anticipated		**18300**
TOTAL INCOME	**21750**	**18300**
	Combined Total: $	**40050**
EXPENSES		
Project Expense Categories	**Cost to Date**	**Anticipated**
Electronics	650	6500
Mechanical	987	5000
Manipulator	400	3800
Science	1468	3000
Software, AI and Autonomy	0	4350
Rover Budget	**3505**	**22650**
Lab Equipment & Facilities	545	2000
Rover Creation	**4050**	**24650**
Backup Parts	200	5000
Travel to Utah	0	40000
Rover Shipping	0	1000
TOTAL EXPENSES	**4250**	**70650**
FORECAST FUNDING SURPLUS		
Current Rover Surplus	**18245**	
Anticipated Rover Surplus	**17400**	
Current Budget Surplus	**17500**	
Anticipated Budget Surplus		**-30600**

Table 1: Project Budget (USD)

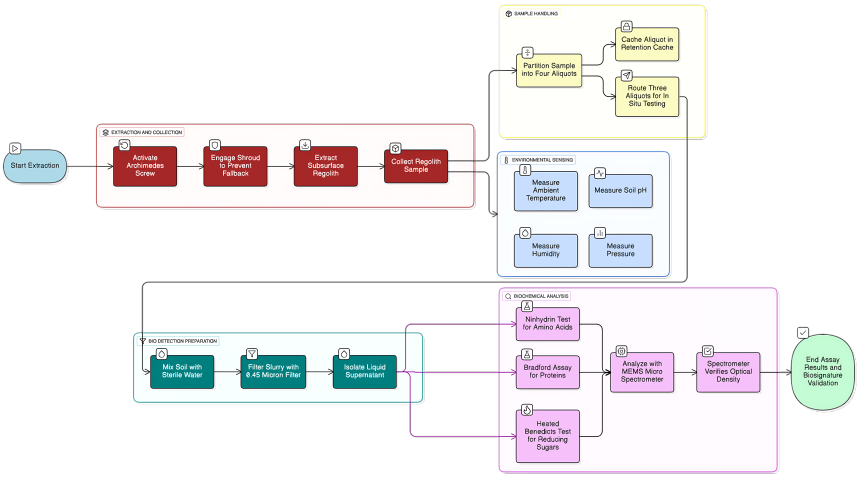

Figure 3: Biosciences Subsystem Workflow Diagram.

Figure 4: Arm of Rover

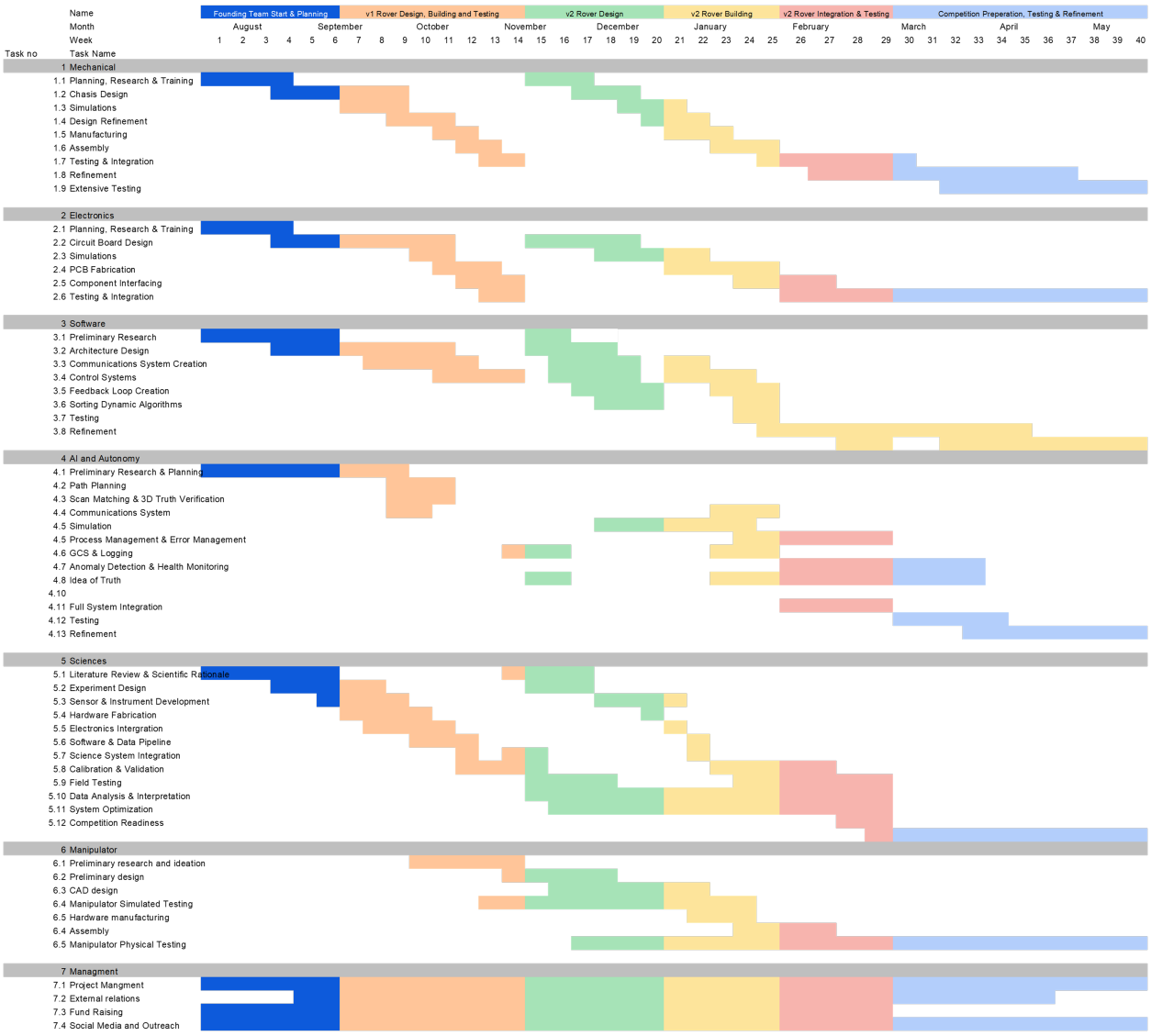

Figure 5: Gantt Chart

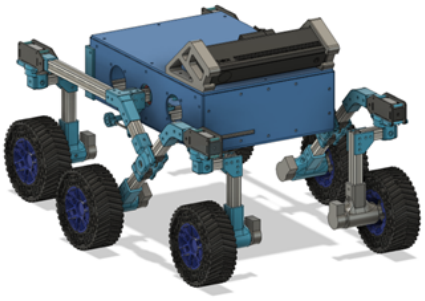

Figure 6: Model of Rover

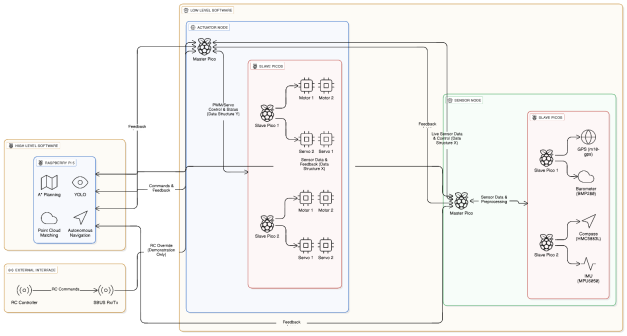

Figure 7: V1 Rover Low-level Software Architecture

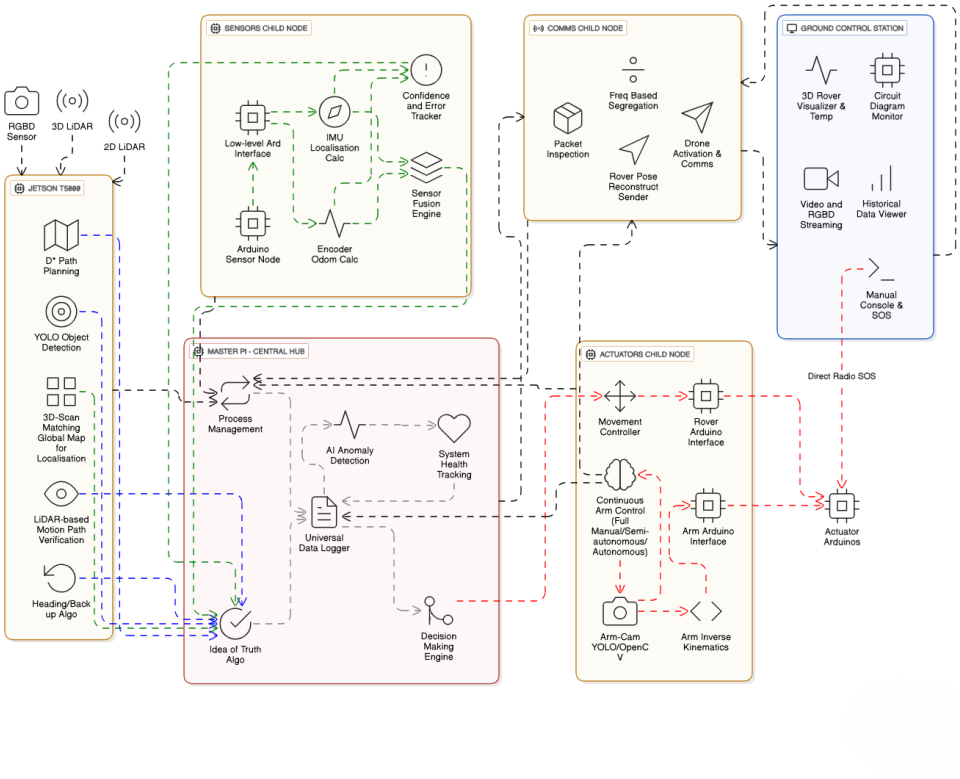

Figure 8: AI and Autonomy Plan for V2 Rover